

Freeing the Trapped Lung

Restoring re-expansion when an organized fibrous rind entraps the lung

Eva Aguirre

MS-MedTech · NC State University & UNC-Chapel Hill



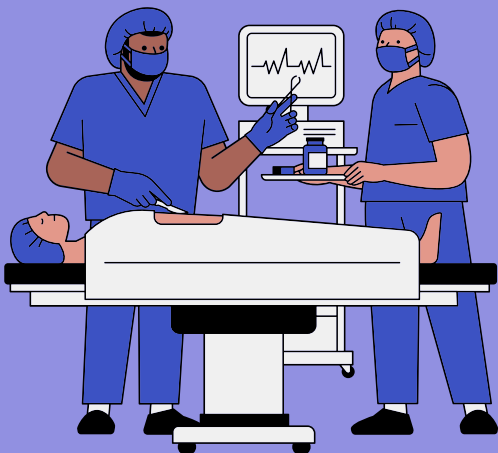
Problem Statement

A way to restore lung re-expansion for patients with a thick pleural rind in order to preserve lung function, avoiding both decortication morbidity and trapped lung disability.

What Decortication Costs

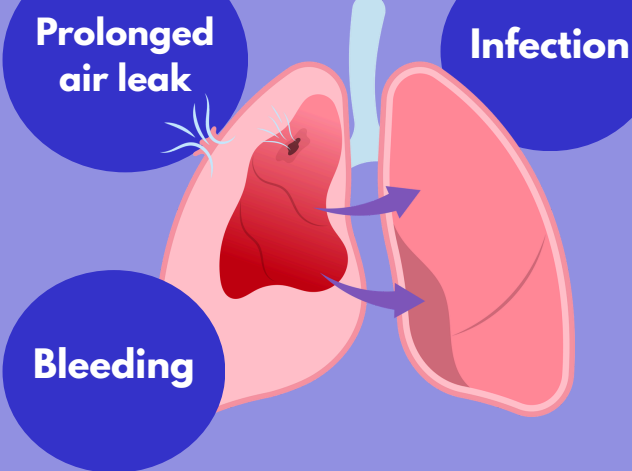
>10,000

decortications performed in the US each year

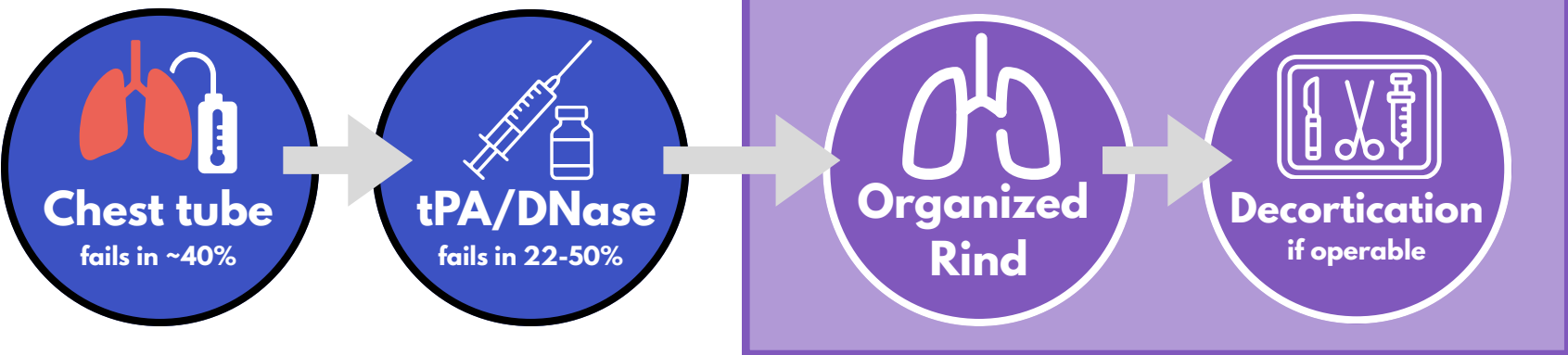


39.3%

complication rate for the only definitive option



Care Pathway

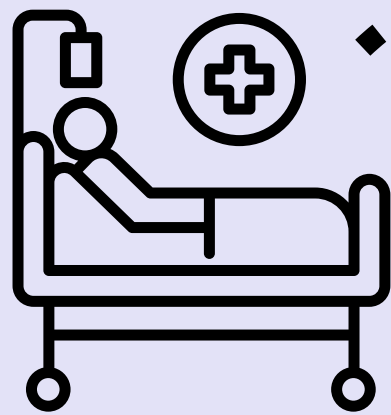
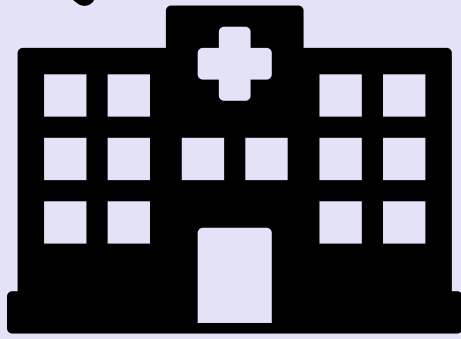


WHERE THE NEED SITS

Current Solutions	Antibiotics + chest tube	Intrapleural tPA/DNase	VATS/open decortication	Indwelling catheter
Acts on organized rind	☐	☐	☒	☐
Avoids an operation	☒	☒	☐	☒
Available to inoperable patients	☒	☒	☐	☒
Restores durable re-expansion	☐	☒	☒	☐

● yes ● partial ○ no

Population



30% require surgery

~1 million hospitalized pneumonia patients

20-40% develop parapneumonic effusions

5-10% progress to empyema

How the Rind Forms



STAGE I
Free-flowing fluid
→ Antibiotics + chest tube



STAGE II
Fibrin, loculation, pus
→ Intrapleural tPA / DNase



STAGE III
Cross-linked collagen peel
→ Surgery only

Why the Drug Fails



tPA → cleaves fibrin



DNase → dissolves the DNA of pus



Neither → touches collagen

Once the matrix is remodelled into collagen, the drug and the target no longer match.

Route A - Biologic

\$42-96M

serviceable market depending on true lytic failure population



An intrapleural collagen-degrading enzyme — no operation

Full Biologic License Application 7-10 years

Prove collagenase can dissolve organized rind



MedTech Innovation + Entrepreneurship

References



"If there were a way to get rid of the rind without surgery, nobody would miss doing decortications."



Thoracic surgeon, on the value of a non-surgical option

Route B - Device

\$5-6M

U.S. decortication market plus ~3M in dense adhesiolysis



Potential 510(k) tool claim 1-3 years Bench or animal data